

Introduction to Internet of Things(IoT)



Introduction

- Security means preventing safeguarding connected device and network in the internet of things (IoT) falling in the hands of unauthorized people.
- The sender of data may wish to be assured that only the intended party actually receives designated data.
- The receiver must authenticate first that is the one who is authorized to receive data and thereafter the sender exchanges data.



Need of Security in IoT

➤ Confidentiality :

- As IoT data travels through multiple hops in a network, a proper **encryption mechanism** is required to ensure the confidentiality of data.

➤ Integrity :

- It refers to the method for ensuring that data is real, accurate and safeguard from **unauthorized use modification**.

➤ Availability :

- It is the guarantee of reliable and constant access to your sensitive data by authorized people. The attacks on IoT devices may block the provision of service through the conventional **denial-of-service attack**.



Cyber Security

- Cyber security deals with the security of information technology. It is also refers to protecting computers, networks, programs and data from unauthorized users.
- Securing your PC
 - Creating strong and Unique Password
 - Keep window and devices Updated
 - Use 2FA (Two Factor Authentication))
 - Install, update and use anti-virus software
 - Encrypt your Connection to the Internet
 - Disable UpnP



Type of Cyber Attacks

- Web Based Attack
- System Based Attack



Web Based Attack

- Injection attacks
- DNS Spoofing
- Session Hijacking
- Phishing
- Brute force



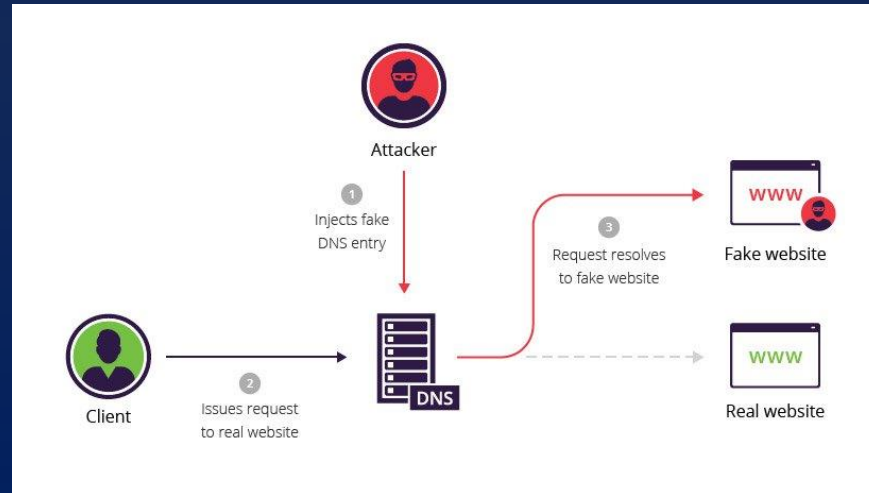
Injection Attacks

- Injection attacks are amongst the oldest and most dangerous web application attacks.
- Injection injects application code which can execute operating system commands as the user running the web application.
- They can result in data theft, data loss, loss of data integrity, denial of Service as well as full system compromise.
- Injection is a major problem in web security.
- Injection attacks, particularly SQL injections (SQLi attacks) and cross-site Scripting (XSS) are the most common injection attacks.



DNS Spoofing

- DNS spoofing, also referred to as DNS cache poisoning, is a form of computer security hacking in which corrupt Domain Name System data is introduced into DNS resolver's cache, causing the name server to return an incorrect result record , e.g., an IP address.
- This result in traffic being diverted to the attacker's computer (or any other computer).



Session Hijacking

- Hijacking take control of various parts of your web browser, including your home page, search page, and search bar.
- They may also redirect you to certain site which is configured to return a malicious IP address.
- Some will even redirect you to their own search engine when you attempt a search.



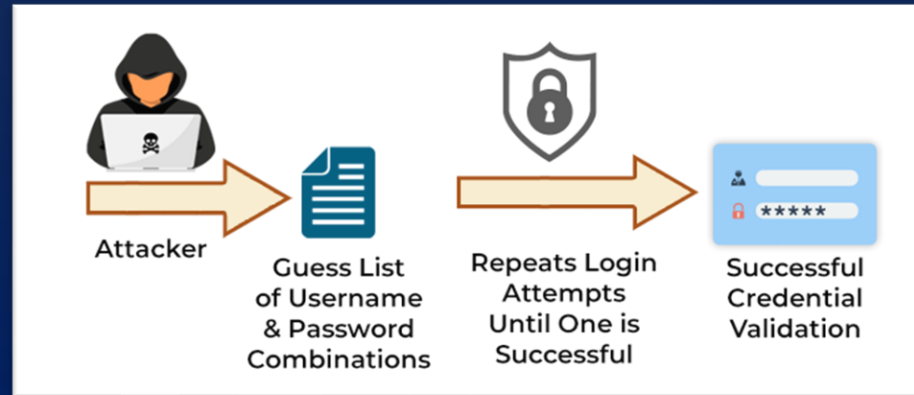
Phishing

- It is an attempt of acquiring sensitive information such as usernames, password, and credit card details, by hackers in an electronic communication.
- Phishing is carried out either by email spoofing (creating of email message with forged sender address) or instant messaging (online chat which offers real-time text transmission over the internet), and it directs users to enter the details at fake website which appears to be identical to the legitimate one.



Brute force

- A brute force attack (also known as brute force cracking) is an attempt to crack a password or user name or find a hidden web page, or find the error approach and hoping, eventually, to guess correctly.
- Attackers let a computer do the work trying different combination of usernames and password, for example until they find one that work.
- The most basic brute force attack is a dictionary attack, where the attacker works through a dictionary of possible password and tries them all.



System Based Attach

- Virus
- Worm
- Trojan horse
- backdoors
- Bots



Virus

- A computer virus is a kind of malware program, which, when executed, gets replicated by inserting copies of itself into another computer programs, data files of the hard drive.
- When the replication process of virus gets succeeded, the affected areas are called as “infected”.



Worm

- A worm is a destructive program that is designed to corrupt database either all at once or over a period of time.
- After the worm code is written, it is hidden with an existing program, and once that program is loaded into the computer, the worm replicate by attaching copies of itself to other program in the system.
- A virus is similar to a worm, which is also a destructive program, however, the virus always attaches itself to other programs, whereas the worm may not. Worm are less harmful than viruses.
- They do not require any carrier program and spread by duplicating themselves.
- They are merely found in the network environment.



Trojan horse

- A Trojan horse is a program that invades a computer system by secretly attaching itself to a valid program download into the computer.
- It may be used to locate password information, or it alter an existing program to make it easier to gain access to it.
- It has no way replicate automatically.



backdoors

- This a program that sends information back to its creator about the infected system, making it easy for that person to hack into the infected system and take control of it or read sensitive data.



Bots

- A bots short for “robot” and also called an internet bot is an automated, means they run according to their instruction without a human user start them up.
- Bots often imitate or replace a human user’s behaviour. Typically they do repetitive tasks, and they can do them much faster than human user could.



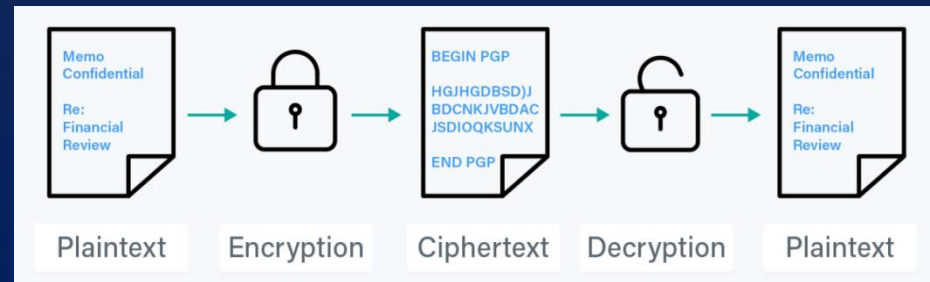
Protecting IoT Devices

- PKI (Public Key Infra-structure) and digital Certificate
- API (Application Performance indicator) security
- Integrating Items
- Identity management
- Hardware Security
- Strong encryption
- Network Security
- Network Access control
- Security Gateway



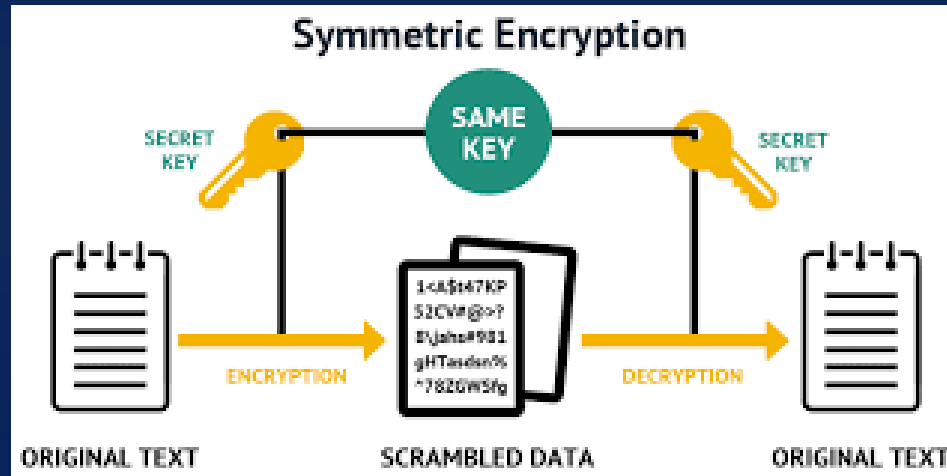
Cryptography

- Cryptography is the science of information security.
- It is closely related to the disciplines of cryptology (it is the mathematics, such as a number theory and the application of formulas and algorithms) and cryptanalysis (it refers to the study of ciphers, cipher text with a view to finding weakness in them that will permit retrieval of plan text from he cipher text, without necessarily knowing the key or the algorithms).
- Cryptography classified into two categories.
 - Symmetric key cryptography
 - Asymmetric key cryptography



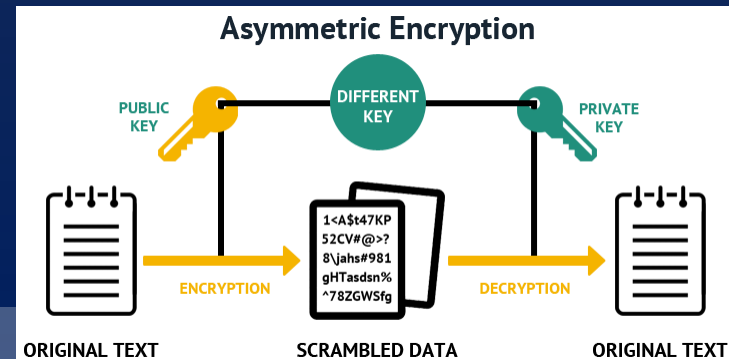
Symmetric key cryptography

- An encryption system in which the sender and receiver of a message share a single, common key that is used to encrypt and decrypt the message.



Asymmetric key cryptography

- It is also known as **public key cryptography**.
- The encryption process where different keys are used for encrypting and decrypting the information.
- Keys are different but are mathematically related, such that retrieving the plain text by decrypting cipher text is feasible.
- Public-key or Asymmetric key encryption algorithms use a pair of keys, a **public key associated with the creator/sender** for encrypting a message and a private key that only the originator knows for decrypting that information.



Artificial Intelligence (AI)

- Artificial Intelligence is one of the emerging technologies that try to simulate human reasoning in AI system.
- You are already using AI in your day-to-day lives for example a virtual assistant such as Siri, Google Assistant or Cortana all are our digital personal assistant.
- AI is vast subject and its field of study includes many major subfields under it.
 - Machine Learning
 - Neural Network
 - Deep Learning
 - Computer Vision
 - Natural language processing



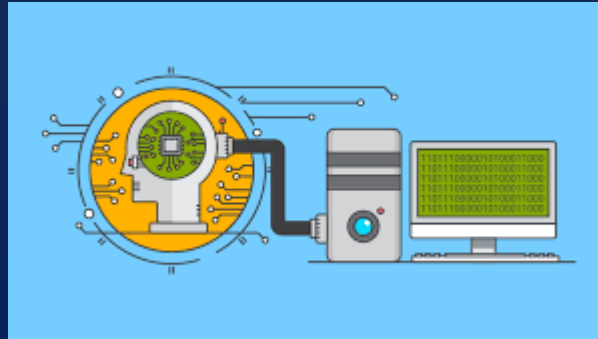
Machine Learning

- Machine learning is an application of AI that provides systems the ability to automatically learn and improve from experience without being explicitly programmed.



Neural Network

- A computer system modelled on the human brain and nervous system.
- These are brain-inspired network of interconnected layers of algorithms, called neurons, that feed data into each other, and which can be trained to carry out specific tasks.



Deep Learning

- A machine learning technique that teaches computer to learn by example.



Computer Vision

- Computer vision system is a technology of obtaining models to control information from visual data.
- For example, a spying aeroplane takes photographs which are used to figure out spatial information or map of the area.



Natural language processing

- It is possible to interact with the computer that understand human language.



Application of AI

- Gaming
- Natural Language Processing
- Expert System
- Vision System
- Speech Recognition
- Handwriting Recognition
- Intelligent Robotics



Advantages of AI

- Reduction in Human Error
- Takes risks instead of Humans
- Available 24x7
- Helping in Repetitive Jobs
- Digital Assistance
- Needs no breaks
- Faster Decisions
- New Inventions



Disadvantages of AI

- High Cost of Creation
- Making humans Lazy
- Unemployment
- No Emotions
- Lacking Out of Box Thinking



THANK YOU

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